



The 66th ASH Annual Meeting Abstracts

POSTER ABSTRACTS

721.ALLOGENEIC TRANSPLANTATION: CONDITIONING REGIMENS, ENGRAFTMENT, AND ACUTE TOXICITIES

Intestinal Microbiota Transplant Prior to Allogeneic Stem Cell Transplant (MAST): A Multi-Center Randomized Double-Blinded Placebo-Controlled Phase IIa Trial

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Background and significance: Reduced gut microbiome diversity pre-allogeneic hematopoietic cell transplantation (HCT) strongly correlates with poorer survival after the procedure. Most hematologic malignancy patients undergoing HCT have previously received intensive chemotherapy, resulting in prolonged neutropenic episodes and opportunistic infections (increasingly with multidrug-resistant organisms (MDRO)), requiring broad-spectrum antibiotics. The combination of chemotherapy and antimicrobial use has particularly been linked to reduced gut microbiome diversity.

Intestinal microbiota transplant (IMT) is a novel treatment approach that can restore this perturbed diversity and has shown promise in a cohort of patients colonized with MDRO in their intestine by reducing post-HCT infection burden and improving overall survival. Given that most patients undergoing HCT will have a disrupted microbiome from prior therapies, we hypothesized that offering IMT prior to initiation of HCT conditioning could improve microbiome diversity during the early stages of HCT, with potential to reduce the frequency of infective complications, and improve outcomes of HCT. Studies using IMT post-HCT have shown variable results, but one distinctive aspect of this study is *pre*-HCT administration of IMT, aiming to 'prehabilitate' the gut prior to the microbiota insults inherent to HCT.

Study Design and Methods: The trial is registered (ClinicalTrials.gov ID: NCT 6355583) and recruitment started in 2024. Fifty adult patients planned to receive allogeneic HCT for hematologic malignancies will be recruited into this phase IIa trial, and randomized 1:1 to receive capsulized IMT (10 capsules of EBX-102-02, a next generation, full-spectrum microbiome product derived from pooled screened donor stool; EnteroBiotix Ltd) or matched placebo at 14 days prior to initiation of HCT conditioning, and followed for up to 12 months. The trial will be performed at seven UK centers.

Patients will be eligible if they have achieved complete remission (<5% blasts), had at least two cycles of intensive chemotherapy, and have received broad-spectrum antibiotics within three months prior to HCT. Exclusion criteria include patients not fit for allogeneic HCT, active infection, chronic intestinal disease, bleeding disorders, allergies to capsule components, difficulties in swallowing, infection with HIV, hepatitis B or C, prior malignancy, administration of IMT within 3 months prior to enrolment, and use of probiotics during after recruitment; patients undergoing umbilical cord transplant are also excluded.

The primary outcome will be to assess the increase in alpha diversity (inverse Simpson's index) between pre-IMT and that measured at 42 days after IMT administration (day +28 of HCT), comparing the difference between patients receiving IMT compared to placebo. Secondary outcomes will include tolerability, the dynamics of gut microbiome diversity metrics and taxonomy over all timepoints assessed, as well as clinical outcomes (including burden of invasive infections, days of fever, admission to intensive care, development of graft-vs-host disease (GvHD), non-relapse mortality, survival, and GvHD-free relapse-free survival). Exploratory outcomes will include the impact of IMT upon neutrophil and platelet engraftment, and T-cell immune reconstitution and repertoire, as well as exploring changes in a range of gut microbial metabolites.

Disclosures Innes: Novartis: Membership on an entity's Board of Directors or advisory committees, Other: Conference fees, Speakers Bureau; Incyte: Speakers Bureau. **Wheeler:** GSK: Current Employment, Current equity holder in publicly-traded company.

<https://doi.org/10.1182/blood-2024-204750>